

# Elements of Statistics and Probability

Nabil Ahmed Uthso

School of Data and Computational Sciences

BRAC University

E-mail: [nabil.ahmed@bracu.ac.bd](mailto:nabil.ahmed@bracu.ac.bd)

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# Random Variables

- A **random variable** is a variable that takes on numerical values as a result of a random experiment or measurement, and associates a probability with each possible outcome

# Random Variables

- Consider the experiment of flipping two fair coins
- The possible outcomes are:  $S = \{HH, HT, TH, TT\}$
- Let  $X$  = the number of heads. So,  $X = \{0, 1, 2\}$
- Then the probabilities associated with each value of  $X$  can be represented by the following table:

| $x$        | 0   | 1   | 2   |
|------------|-----|-----|-----|
| $P(X = x)$ | 1/4 | 2/4 | 1/4 |

# Types of Random Variable

- 1 Discrete random variable
- 2 Continuous random variable

# Types of Random Variable

- **Discrete random variable:** A **discrete** random variable is a random variable whose possible values either constitute a finite set of values or an infinite sequence of numbers that is a countably infinite set of numbers. Example:
  - The number of cars crossing an intersection every hour
  - The number of phone calls received per day at a call centre
- **Continuous random variable:** **Continuous** random variables can represent any value within a specified range or interval and can take on an infinite number of possible values. Example:
  - The time taken to serve a customer at a call centre
  - The daily temperature at noon

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# Probability Mass Function (PMF)

- The probability distribution of a **discrete** random variable is known as **Probability Mass Function (PMF)**
- If  $X$  is a discrete random variable with possible values  $x_1, x_2, \dots, x_n$ , where each value has a corresponding probability  $P(X = x_i); i = 1, 2, \dots, n$ , the probability mass function  $f(x)$  of  $X$  is defined by

$$f(x) = \begin{cases} P(X = x_i); & \text{if } X = x_i, i = 1, 2, \dots, n \\ 0; & \text{otherwise} \end{cases}$$

- And has the following properties:
  - $0 \leq f(x_i) \leq 1$  for all  $x_i$
  - $\sum_{i=1}^n f(x_i) = 1$

# Example

- Consider the experiment of flipping two fair coins
- Let  $X$  = The number of heads. So,  $X = \{0, 1, 2\}$

$$f(x) = \begin{cases} P(X = x_i); & x_i = 0, 1, 2 \\ 0; & \text{otherwise} \end{cases}$$

| $x_i$    | 0   | 1   | 2   |
|----------|-----|-----|-----|
| $f(x_i)$ | 1/4 | 2/4 | 1/4 |

## Example 2

- Let  $X$  be a random variable with probability function defined as follows:

|          |      |      |      |      |
|----------|------|------|------|------|
| $x_i$    | 2    | 4    | 6    | 8    |
| $f(x_i)$ | 2/10 | 1/10 | 4/10 | 3/10 |

- $P(X = 4) = \frac{1}{10}$
- $P(X > 4) = P(X = 6) + P(X = 8) = \frac{4}{10} + \frac{3}{10} = \frac{7}{10}$
- $P(X \leq 6) = P(X = 2) + P(X = 4) + P(X = 6) = \frac{2}{10} + \frac{1}{10} + \frac{4}{10} = \frac{7}{10}$
- $P(2 < X \leq 8) = P(X = 4) + P(X = 6) + P(X = 8) = \frac{1}{10} + \frac{4}{10} + \frac{3}{10} = \frac{8}{10}$
- $P(X = 3) = 0$

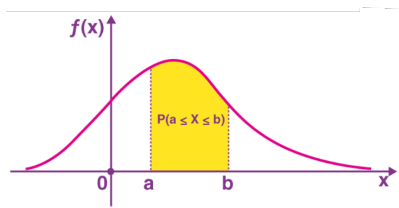
# Probability Density Function (PDF)

- The probability distribution of a **continuous** random variable is known as continuous probability distribution/ **Probability Density Function (PDF)**
- If  $X$  is a continuous random variable, the probability density function  $f(x)$  of  $X$  is a function such that for any two numbers  $a$  and  $b$  with  $a \leq b$

$$P(a \leq X \leq b) = \int_a^b f(x)dx$$

# Probability Density Function (PDF)

- That is, the probability that  $X$  takes on a value in the interval  $[a, b]$  is equivalent to the area below the graph of  $f(x)$  between the interval  $[a, b]$ . The graph of  $f(x)$  is often referred to as the density curve



# Probability Density Function (PDF)

- A valid PDF  $f(x)$  has the following properties:
  - $f(x) \geq 0$
  - $\int_{-\infty}^{\infty} f(x)dx = 1$
- **Note:** The value of  $P(x)$  for any point value of  $X$ , say  $X = k$ , will always be 0. That is,

$$P(X = k) = 0; k \in \mathbb{X}$$

# Example

- The probability density function of a random variable  $X$  is defined as

$$f(x) = \begin{cases} x; & 0 \leq x < 1 \\ 2 - x; & 1 \leq x < 2 \\ 0; & x \geq 2 \end{cases}$$

- (a) Verify that it is a pdf.
- (b) Find  $P(0.5 \leq X \leq 1.5)$ .

# Solution

(a)

$$\begin{aligned}\int_{-\infty}^{\infty} f(x)dx &= \int_0^1 xdx + \int_1^2 (2-x)dx + \int_2^{\infty} 0dx \\ &= \left[\frac{x^2}{2}\right]_0^1 + \left[2x - \frac{x^2}{2}\right]_1^2 \\ &= 1\end{aligned}$$

# Solution

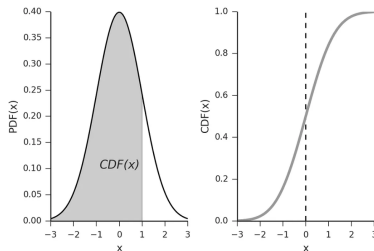
(b)

$$\begin{aligned}P(0.5 \leq X \leq 1.5) &= \int_{0.5}^1 x dx + \int_1^{1.5} (2 - x) dx \\&= \left[ \frac{x^2}{2} \right]_{0.5}^1 + \left[ 2x - \frac{x^2}{2} \right]_1^{1.5} \\&= \frac{3}{4}\end{aligned}$$

# Cumulative distribution function (CDF)

- The CDF of a random variable  $X$  is denoted by  $F(x)$  is defined as

$$F(x) = P(X \leq x) = \int_{-\infty}^x f(x)dx$$



# Example

- Find the cdf of the pdf of the  $X$ :

$$f(x) = 2x; 0 < x < 1$$

- CDF:

$$F(x) = \int_{-\infty}^x f(x)dx = \int_0^x 2xdx = x^2$$

- Hence the cdf can be written as

$$F(x) = \begin{cases} x^2; & 0 < x < 1 \\ 1; & x \geq 1 \end{cases}$$

# Exercise

- Find the cdf of the pdf of  $X$ :

$$f(x) = \begin{cases} x; & 0 < x \leq 1 \\ 2 - x; & 1 < x \leq 2 \\ 0; & \text{otherwise} \end{cases}$$

# CDF to PDF

- CDF:  $F(x)$
- PDF:  $f(x) = \frac{d}{dx}F(x)$

# Example

- Find the PDF of the CDF below:

$$F(x) = \begin{cases} x^2; & 0 < x < 1 \\ 1; & x \geq 1 \end{cases}$$

- PDF:

$$f(x) = \begin{cases} 2x; & 0 < x < 1 \\ 0; & \text{otherwise} \end{cases}$$

# Thank You